Chinyere E. Ugwuanyi

New Brunswick, NJ | (304)-305-0259 | chinyere.ugwuanyie@gmail.com | www.linkedin.com/in/chinyere-ugwuanyi | https://github.com/ChinyereEU

EDUCATION

Rutgers University – MBS, Analytics & Data Science | GPA: 3.89 | *May 2026*

- **Relevant Courses**: Regression, BI & Visual Analytics, Data Warehousing, Data Management, Data Structures (Java)

West Virginia State University – B.S. Chemistry | GPA: 3.84 | *April 2021*

TECHNICAL SKILLS

- **Programming & Tools:** Python (Pandas, NumPy, Scikit-learn), R, SQL/MySQL, SPSS, Qualtrics, Tableau, Excel
- **ML & Statistics:** Regression, Classification, Survival Analysis, Reliability Testing, Experimental Design, Longitudinal and Cohort Analysis
- **Data Engineering & Validation:** Data Cleaning, Feature Engineering, Model Validation, Quality Assurance

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | *Rutgers - Office of Institutional Research & Decision Support (OIRDS)* *March 2025 – Present*

- Led multi-campus administration of the Rutgers Undergraduate Experience Survey, managing project setup, distribution logic, branding, and response tracking in Qualtrics (n = **40,000+ students**).
- Integrated and cleaned multi-source data by combining survey responses with 941 official student organization records extracted via university Engage API using R; developed text standardization, fuzzy-matching, and duplicate resolution workflows in R to resolve unstructured inputs and ensure accurate entity linkage for reporting and incentive allocation.
- Executed large-scale roster QA and validation for seed roster files using R, Python, SPSS, and Excel to ensure accurate survey deployment.
- Conducted **longitudinal cohort analyses** on student engagement, retention, and equity gaps to inform student success planning and strategic initiatives.
- Developed dashboards and executive-level reports to translate survey and institutional data into actionable insights for administrators and academic stakeholders.
- Collaborated with academic units and student affairs departments to refine research questions, align analyses with institutional priorities, and support data-informed decision-making.
- Developed an M-index dining utilization algorithm (n = **4,800+ students**, $p < 0.001$) supporting evidence-based resource allocation.

Student Researcher - Computer Science Education | *Rutgers Women in Computer Science* *Spring 2024 - Fall 2024*

- Designed Qualtrics surveys (n = **900+ students**) to assess Computer Science and IT&I curriculum effectiveness and student engagement.
- Built visualizations using Tableau and Plotly to support curriculum improvement recommendations.

Treasurer & At-Large Member | *RUSA Allocations Board - Rutgers University* *Fall 2024*

- Supported equitable distribution of $1M+ funds to **400+** student clubs; tracked budgets, allocation requests, and bi-weekly appeals processes.

PROJECTS

ZIP-Code-Level COVID-19 Early Detection & Surveillance Model (NJ) *(In Progress)*

- Engineer dynamic spread indicators (velocity, acceleration, jerk, rolling variance, volatility ratios, correlation) from daily ZIP-code-level COVID-19 case data to capture real-time transmission dynamics.
- Apply drift detection (KS test, CUSUM, Isolation Forest) to identify structural shifts and emerging outbreaks earlier than traditional county-level models; benchmark performance against established epidemiological approaches to improve targeted intervention strategies.

WiDS Datathon 2026 - Wildfire Survival Modeling Project *(In Progress)*

- Build discrete-time survival model predicting when wildfires will reach within 5km of evacuation zones; (risk horizons: 12h, 24h, 48h, 72h) to support emergency-response logistics.
- Expand events to person-period data, engineer time encodings, train gradient-boosted hazard model, evaluate with cross-validated Brier scores.
- Produce calibrated, time-specific risk curves to inform evacuation timing, resource staging, and operational planning under uncertainty.

Fake News Categorization – ML Pipeline (~45,000 articles) GitHub Repo

- Built a text classification pipeline using Python, TF-IDF, cosine similarity, and Nearest Neighbors to detect misinformation and reduce training bias and data leakage.
- Engineered 20+ linguistic features (e.g., punctuation density, sentence structure, readability scores) supporting interpretable modeling.

Airbnb NYC Price Prediction Model – Applied ML GitHub Repo

- Built and validated regression model (40K+ listings, 5-fold CV) using Python, identifying key drivers of NYC price variability.

CERTIFICATIONS / PRESENTATIONS / ADDITIONAL INFORMATION

- **American Educational Research Association (AERA) Annual Meeting** (2026) – Co-author
  From Survey to Success: Predicting First-Year Retention Through Longitudinal Data Analysis
- **Rutgers University – Office of Institutional Research & Decision Support** - New Brunswick, NJ | *December 2025*
  Meet the Cohort of 2025: What Multi-Year New Student Survey Data Reveals About Our Newest Scarlet Knights
- **Cornell Tech** | **Break Through Tech AI/ML Program** - Machine Learning Foundations | *July 2025*
- **CITI Program** | **Social / Behavioral / Epidemiologic Research Investigators** | *June 2024*